

Disaster Risk Management System

Architecture of RESQGUIDE:

Layers included:

1. Frontend (React + UI Pages)

- Login
- Register
- Building Details Form
- Safety Measures
- Risk Result
- Alerts & Safe Places

2. Backend (Node.js + Express)

- Authentication (JWT)
- Password Hashing
- Rule-based Risk Evaluation
- API Handling

3. Databases

- PostgreSQL
 - Users
 - Buildings
 - Safety Measures
 - Risk Assessment
- MongoDB
 - Disaster Alerts
 - Safety Instructions
 - Safe Places

4.External APIs

Weather / Disaster API

- Google Maps API

Flow:

- User submits building and safety data
- Backend evaluates risk using predefined rules
- Weather API triggers disaster alerts
- Google Maps API identifies nearest safe places
- Results displayed to user in real time.

PostgreSQL :

users TABLE:

Column Name	Data Type	Key / Constraint	Example Value
user_id	SERIAL	Primary Key	101
name	VARCHAR(100)	NOT NULL	Ravi Kumar
email	VARCHAR(150)	UNIQUE	ravi@gmail.com
password_hash	TEXT	NOT NULL	\$2b\$10\$abc123
role	VARCHAR(20)	DEFAULT user	admin
created_at	TIMESTAMP	Auto Generated	2026-01-26

buildings TABLE:

Column Name	Data Type	Key / Constraint	Example Value
building_id	SERIAL	Primary Key	201

Column Name	Data Type	Key / Constraint	Example Value
user_id	INTEGER	Foreign Key	101
building_type	VARCHAR(50)	NOT NULL	Factory
floors	INTEGER	NOT NULL	4
construction_year	INTEGER	NOT NULL	2015
latitude	DECIMAL(9,6)	NOT NULL	16.306700
longitude	DECIMAL(9,6)	NOT NULL	80.436500
created_at	TIMESTAMP	Auto Generated	2026-01-26

safety_measures TABLE:

Column Name	Data Type	Key / Constraint	Example Value
safety_id	SERIAL	Primary Key	301
building_id	INTEGER	Foreign Key	201
fire_extinguisher	BOOLEAN	Default FALSE	TRUE
emergency_exit	BOOLEAN	Default FALSE	TRUE
cctv	BOOLEAN	Default FALSE	FALSE
medical_kit	BOOLEAN	Default FALSE	TRUE
structural_strength	VARCHAR(20)	NOT NULL	High

Risk assessment TABLE:

Column Name	Data Type	Key / Constraint	Example Value
risk_id	SERIAL	Primary Key	401
building_id	INTEGER	Foreign Key	201
risk_score	INTEGER	NOT NULL	68
risk_level	VARCHAR(20)	NOT NULL	High
assessment_date	TIMESTAMP	Auto Generated	2026-01-26

MongoDB :**disaster alerts COLLECTION:**

Field Name	Data Type	Description	Example Value
_id	ObjectId	Unique document ID	65af12...
disaster_type	String	Type of disaster	Flood
severity	String	Alert level	High
location	String	Affected area	Guntur
alert_message	String	Alert text	Flood warning issued
timestamp	Date	Alert time	2026-01-26

safety recommendations collection:

Field Name	Data Type	Description	Example Value
_id	ObjectId	Unique document ID	65af23...
disaster_type	String	Disaster category	Flood
risk_level	String	Risk severity	High
instructions	Array	Safety steps	Move to higher ground
last_updated	Date	Update time	2026-01-26

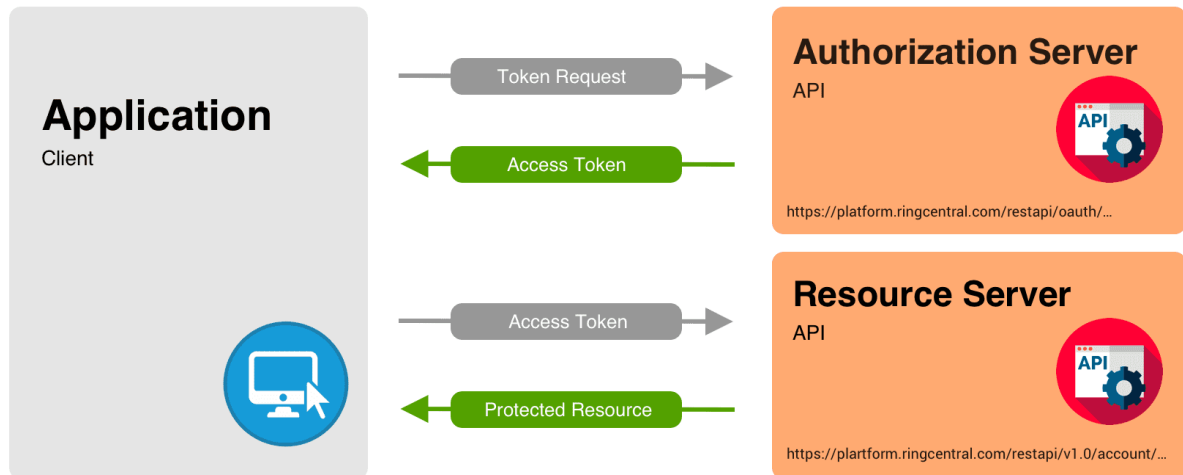
safe places collection:

Field Name	Data Type	Description	Example Value
_id	ObjectId	Unique document ID	65af34...
place_name	String	Safe location name	Govt High School
place_type	String	Shelter type	School
latitude	Number	Latitude	16.308900
longitude	Number	Longitude	80.439800
capacity	Number	Max people	500
supported_disasters	Array	Disaster types	Flood, Cyclone

Security tokens:

JWT Token authentication:

Token Flow in ResGuide:



Password Hashing:

